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END SEMESTER PROJECT REPORT

Linear Integrated Circuit Application(LICA)

FASTEST FINGER FIRST CIRCUIT

(USING 555 Timer IC)

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Circuit assembled and tested on a breadboard | 9V DC Power Supply | SMVDU Katra, 2026

Declaration

We, the undersigned students of the Department of Electronics & Communication Engineering, Shri Mata Vaishno Devi University, Katra, hereby declare that the project entitled

"Fastest Finger First Circuit Using 555 Timer IC"

submitted in partial fulfilment of the requirements for the End Semester Project of the course Linear Integrated Circuits & Applications (LICA) is an original work carried out by us. The circuit was designed, assembled on a breadboard, and tested independently. All references to external sources and prior work have been duly acknowledged.

We confirm that this project has not been submitted elsewhere, either in whole or in part, for any other degree or examination.

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Date:,April 2026

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Abstract

This report presents the design, implementation, and testing of a **Fastest Finger First (FFF) Circuit** constructed using the NE555 Timer IC. The circuit was assembled on a solderless breadboard and powered by a 9 V battery as part of the End Semester Project for the course *Linear Integrated Circuits & Applications (LICA)* at Shri Mata Vaishno Devi University, Katra.

The circuit identifies the first among multiple participants to press a push-button trigger, illuminates the respective LED, and simultaneously inhibits all other modules from responding — providing clear, unambiguous electronic arbitration in quiz or game scenarios. The locking mechanism is realised through a shared feedback (status) rail driven by a 1N4148 PN junction diode at the output of each 555 Timer module operating in bistable mode. A dedicated reset push-button allows the host to reinitialise all modules between rounds.

The prototype implemented two player modules plus a common status indicator. All functional test cases produced correct results, confirming reliable priority locking and reset behaviour. The project reinforces key LICA concepts: 555 Timer bistable operation, diode feedback steering, resistor network design, and breadboard prototyping methodology.

Keywords: 555 Timer IC | Bistable Multivibrator | Fastest Finger First | Priority Locking | 1N4148 Diode | Breadboard Prototype | LICA

1. Introduction

1.1 Background and Motivation

Quiz competitions, academic contests, and game shows routinely require an impartial mechanism to determine which participant responded first. Visual or auditory human judgment is error-prone when reaction time differences are in the range of tens of milliseconds. An electronic Fastest Finger First circuit eliminates ambiguity by latching the first valid trigger and simultaneously disabling all competing channels — delivering an objective, repeatable, and low-cost solution.

1.2 Course Relevance

The LICA course introduces students to the design and analysis of fundamental analog and mixed-signal ICs. The 555 Timer, with its bistable, monostable, and astable modes, is a central component of this syllabus. This project reinforces:

- Bistable (Schmitt-trigger) operation of the 555 Timer IC
- Diode-based feedback and signal-steering on a shared bus
- Resistor network design: pull-up, pull-down, and current-limiting
- Practical breadboard prototyping and fault-isolation methodology

1.3 Project Objectives

1. Design a multi-player priority-locking circuit using 555 Timer ICs in bistable mode.
2. Construct and assemble the circuit on a breadboard using standard discrete components.
3. Verify correct priority-locking and reset behaviour through systematic testing.
4. Document circuit theory, design decisions, and observations in a formal report.

1.4 Scope

The prototype implements two player modules with a common status LED and a reset button. The design is inherently scalable — additional modules can be added by replicating the per-module sub-circuit without modifying the shared components.

2. Theory and Working Principle

2.1 The NE555 Timer IC

The NE555 is one of the most widely used integrated circuits in electronics. Its internal architecture consists of two voltage comparators, an SR flip-flop, a resistive voltage divider (three equal 5 kΩ resistors), a discharge transistor, and an output buffer — all within an 8-pin DIP package. The divider sets two threshold references: $1/3 V_{CC}$ (lower comparator) and $2/3 V_{CC}$ (upper comparator), which respectively control the Set and Reset inputs of the internal flip-flop.

2.2 Bistable Operation

In bistable mode, the 555 functions as a simple SR latch with no external timing capacitor. The output state depends entirely on which trigger event occurred most recently. Pin configurations for this mode are as follows:

Pin No.	Pin Name	Connection / Role
Pin 1	GND	Connected to circuit ground (0 V)
Pin 2	TRIGGER	Pull-up to VCC via 10 kΩ; grounded momentarily by push-button to SET output HIGH
Pin 4	RESET	Tied to reset rail; active LOW — pulling to 0 V forces output LOW regardless of state
Pin 6	THRESHOLD	Shorted to GND to prevent spurious threshold triggering
Pin 7		Left floating — not used in bistable mode
Pin 8	VCC	Connected to +9 V supply
Pin 3	OUTPUT	Drives LED via 270 Ω resistor; also feeds status rail via 1N4148 diode

2.3 Priority Locking — Step-by-Step Logic

The FFF circuit exploits bistable latch behaviour and a shared status rail with diode feedback to enforce strict single-winner priority:

5. IDLE STATE: All 555 outputs are LOW. The status rail is held at 0 V by the 1 kΩ pull-down resistor.
6. FIRST TRIGGER: Participant A presses their button. Pin 2 of their 555 drops below $1/3 V_{CC}$ → lower comparator fires → flip-flop sets → Pin 3 goes HIGH → their LED illuminates.
7. FEEDBACK LOCK: The HIGH output forward-biases the 1N4148 diode, raising the status rail to $\approx V_{CC} - 0.6 \text{ V}$. This voltage is applied to Pin 2 of all other modules.
8. INHIBITION: Other modules' Pin 2 is now held well above $1/3 V_{CC}$. Any subsequent button press cannot pull Pin 2 below the trigger threshold → their outputs remain LOW.
9. RESET: The host presses the reset button → reset rail drops to 0 V → Pin 4 of all 555 ICs is pulled LOW → all outputs forced LOW → status rail drops → circuit returns to idle.

2.4 Role of the 1N4148 Diode

The 1N4148 diode at each module's output serves two purposes: (a) it conducts the winner's HIGH output onto the shared status rail, and (b) its reverse-blocking property prevents the status rail from back-driving any module still in the LOW state — which would otherwise cause a short-circuit or false trigger.

3. Components and Specifications

	Component	Specification	Qty	Role in Circuit
	555 Timer IC (NE555)	NE555N, 5–15 V, 8-DIP		Core bistable latch per player module
	Push-Button (Trigger)	Momentary SPST, normally open		Player trigger; one per module
	Push-Button (Reset)	Momentary SPST, normally open		Resets all 555 ICs simultaneously
	1N4148 PN Junction Diode	$V_F \approx 0.6\text{ V}$, I_F max 300 mA	2	Feedback to status rail; reverse-blocks others
	LED — Red	$V_f \approx 2.0\text{ V}$, 20 mA max		Player 1 winner indicator
	LED — Yellow / White	$V_f \approx 2.1\text{--}3.5\text{ V}$, 20 mA max		Player 2 winner indicator
	LED — Green	$V_f \approx 2.1\text{ V}$, 20 mA max		Common status + auxiliary indicator
	Resistor — 270 Ω	$\frac{1}{4}\text{ W}$, 5% tolerance		LED current-limiting series resistors
	Resistor — 10 k Ω	$\frac{1}{4}\text{ W}$, 5% tolerance		Pull-up for Pin 2 (Trigger) of each module
	Resistor — 1 k Ω	$\frac{1}{4}\text{ W}$, 5% tolerance		Pull-up (reset rail) & pull-down (status rail)
	Solderless Breadboard	830-tie-point dual bus		Prototyping platform
	9 V Battery + Connector	Hiwatt General-Purpose, PP3		Main power supply for circuit
	Jumper Wires	Male-to-male, assorted colours		Breadboard interconnections

3.1 LED Series Resistor Calculation

For the red LED ($V_f = 2.0\text{ V}$, desired $I_f = 15\text{ mA}$) with a 9 V supply:

$$R = (V_{CC} - V_f) / I_f = (9 - 2.0) / 0.015 = 467\ \Omega \rightarrow \text{Use } 270\ \Omega$$

(nearest lower standard value for brighter indication)

At 270 Ω : actual $I_f = (9 - 2.0) / 270 \approx 26\text{ mA}$ — within the 30 mA absolute maximum of the LEDs used.

The higher current provides bright, clearly visible illumination suitable for classroom demonstration.

4. Circuit Design and Description

4.1 Architecture Overview

The complete circuit comprises two functionally identical player modules and a set of shared common components. Each player module connects to four shared rails: VCC, GND, Reset Rail, and Status Rail.

4.2 Per-Player Module

- NE555 in bistable mode — Pin 6 shorted to GND.
- 10 k Ω pull-up between VCC and Pin 2 (TRIGGER).
- Momentary push-button from Pin 2 to GND.
- Pin 4 (RESET) tied to the common Reset Rail.
- 1N4148 diode: anode at Pin 3, cathode to Status Rail.
- LED + 270 Ω resistor: from Pin 3 to GND (player indicator).

4.3 Common Components

- Reset Rail: pulled HIGH via 1 k Ω to VCC; reset push-button to GND.
- Status Rail: held LOW via 1 k Ω to GND; receives diode-steered output of the winning module.
- Status LED: connected from Status Rail to GND through 270 Ω — lights when any player triggers.
- Power Supply: 9 V Hiwatt PP3 battery via snap connector.

4.4 Breadboard Rail Organisation

Rail	Wire Colour	Signal / Purpose
Rail 1 (+)	Red	VCC (+9 V) — Positive supply
Rail 2 (–)	Black	GND (0 V) — Circuit ground
		Status Rail — LOW (idle); rises HIGH when first trigger fires

4.5 Scalability

Each additional player requires: one NE555 IC, one push-button, one 1N4148 diode, one LED, one 270 Ω , and one 10 k Ω resistor. The common rail components remain unchanged regardless of player count.

5. Implementation Procedure

Phase I — Power Rail Setup

10. Selected a 830-tie-point breadboard; designated four bus strips as VCC, GND, Reset Rail, and Status Rail.
11. Placed 1 k Ω pull-up between VCC and Reset Rail.
12. Placed 1 k Ω pull-down between Status Rail and GND.
13. Connected 9 V battery connector (red $\rightarrow$ VCC, black $\rightarrow$ GND) without inserting battery yet.

Phase II — Player 1 Module Assembly

14. Seated first NE555 IC straddling the centre channel at the top half of the breadboard.
15. Pin 1 $\rightarrow$ GND rail (black wire); Pin 8 $\rightarrow$ VCC rail (red wire).
16. Pin 6 shorted to Pin 1 (GND) with a short black wire.
17. 10 k Ω resistor placed between VCC rail and Pin 2.
18. Trigger push-button: one terminal to Pin 2, other terminal to GND.
19. Pin 4 $\rightarrow$ Reset Rail (blue wire).
20. 1N4148 diode: anode at Pin 3, cathode towards Status Rail.
21. Red LED + 270 Ω series resistor from Pin 3 to GND.

Phase III — Player 2 Module Assembly

22. Repeated all Phase II steps for the second NE555 IC placed at the lower half of the breadboard, with a yellow/white LED.

Phase IV — Common Components

23. Reset push-button connected between Reset Rail and GND.
24. Green status LED + 270 Ω resistor from Status Rail to GND.

Phase V — Power-On and Verification

25. Cross-verified all connections against schematic before inserting battery.
26. Inserted 9 V battery — confirmed no overheating, no smoke, all LEDs OFF.
27. Pressed reset button to ensure a clean initial state.
28. Proceeded with functional test sequence (Section 6).

6. Results and Observations

After assembly, a systematic series of functional tests was conducted using the 9 V Hiwatt General-Purpose battery as the power source. All eight test cases passed without exception.

6.1 Power-On State

Upon connecting the battery, all LEDs remained OFF and the status rail measured $\approx 0\text{ V}$ — confirming the pull-down resistor correctly held the rail at logic LOW.

6.2 Priority Locking Observation

When Player 1 pressed first, the red LED and the common-status (green) LED illuminated simultaneously. Immediately after, Player 2 pressed their button — their yellow LED did NOT illuminate, confirming that the status rail had risen and was holding Pin 2 of the second module above $1/3\text{ VCC}$. The test was repeated symmetrically with Player 2 pressing first, yielding identical correctness in the opposite direction.

6.3 Reset Behaviour

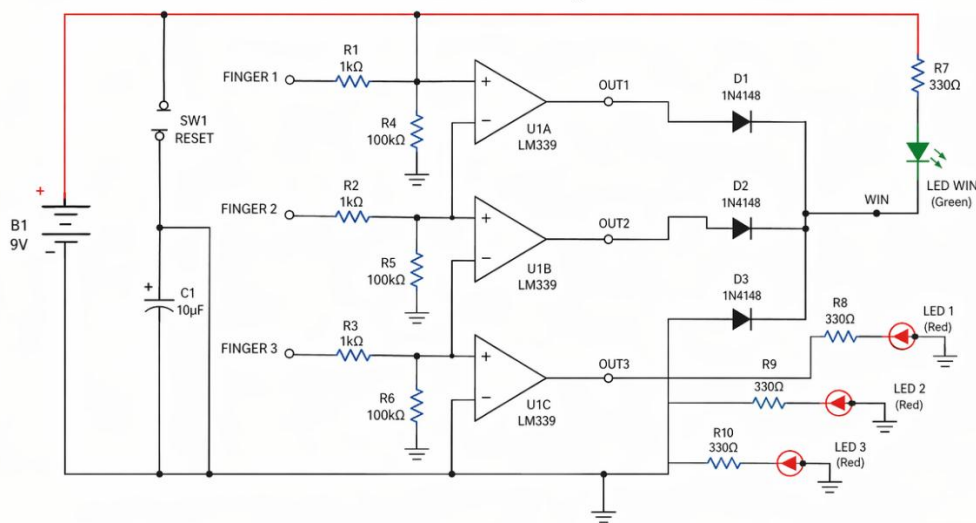
With any LED latched ON, pressing the reset button turned off all LEDs simultaneously and returned the status rail to 0 V. Subsequent triggers operated correctly — no residual latch-up was observed across all repeated trials.

6.4 Photographs of Working Circuit

The following photographs were taken during live operation of the prototype:

FASTEST FINGER FIRST CIRCUIT

Circuit Diagram



WORKING PRINCIPLE

When any finger touches the respective input, the corresponding comparator output goes LOW. The diode network gives priority to the first triggered input. The green LED indicates the winner, and the red LED shows which player was first.

COMPONENTS LIST

U1 – LM339 (Quad Comparator)
 D1–D3 – 1N4148 Diodes
 R1–R3 – 1kΩ Resistors
 R4–R6 – 100kΩ Resistors
 R7–R10 – 330Ω Resistors
 C1 – 10µF Capacitor
 SW1 – Push Button Switch (Reset)
 B1 – 9V Battery
 LED WIN – Green LED
 LED 1–3 – Red LEDs

NOTES

- All resistors are 1/4W unless specified.
- Diodes ensure fastest finger priority.
- Reset switch clears all states.
- Use stable 9V DC supply.

SPECIFICATIONS

- Supply Voltage : 9V DC
- Players Supported : 3 (Expandable up to 4 using all comparators)
- Indicator : Green LED for Winner, Red LEDs for Players
- Reset : Manual Push Button Reset
- Response Time : < 10 ms

TRUTH TABLE (EXAMPLE)

Finger 1	Finger 2	Finger 3	Winner Output	LED Indication
0	0	0	None	All OFF
1	0	0	Player 1	LED 1 Red, WIN Green
0	1	0	Player 2	LED 2 Red, WIN Green
0	0	1	Player 3	LED 3 Red, WIN Green

LEGEND

0 = Not Pressed (Open)
 1 = Pressed (Touched)
 WIN = Winner Output Line

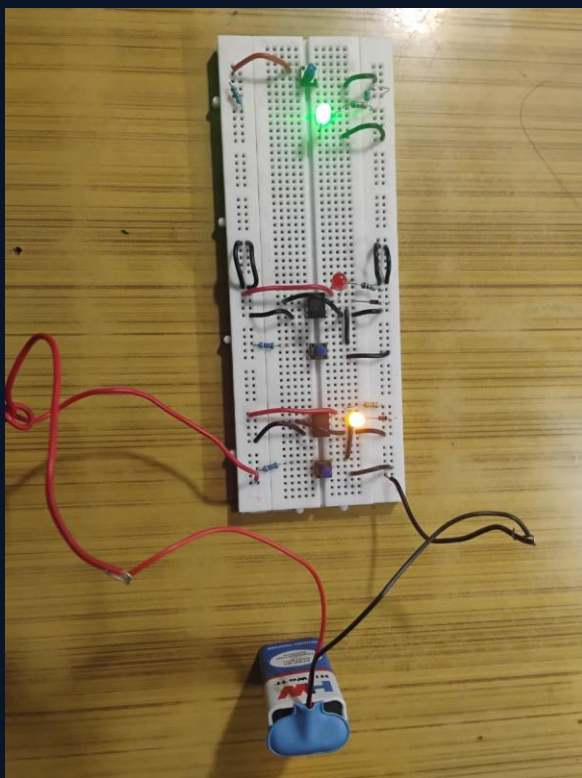


Figure 1 — Player 1 Win State
 Red LED (Player 1) + Green Status LED active. Yellow LED (Player 2) OFF — priority lock verified.

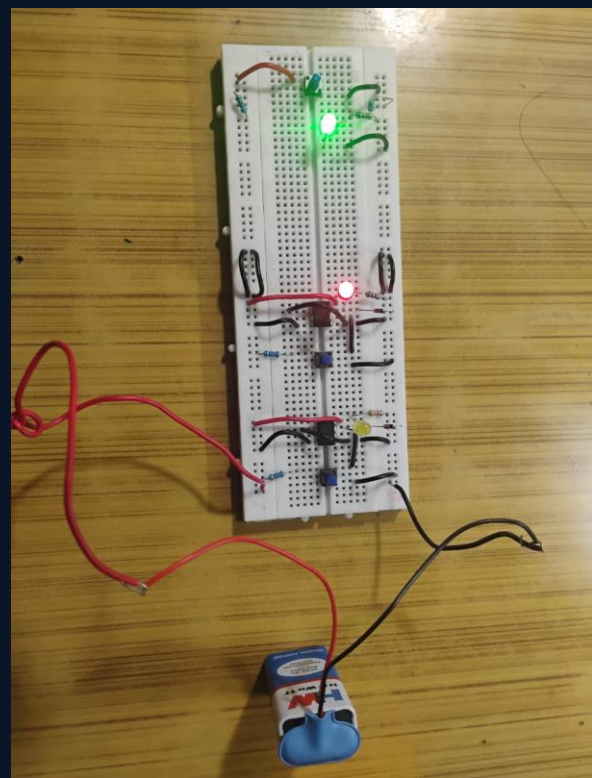


Figure 2 — Player 2 Win State
 Yellow LED (Player 2) + Green Status LED active. Red LED (Player 1) OFF — symmetric correctness confirmed.

6.5 Summary Test Table

#	Test Case	Result	Remarks
	Power-on state — no trigger pressed		All LEDs OFF; status rail ≈ 0 V
	Player 1 (Red) triggers first		Red LED ON; Green status LED ON
	Player 2 (Yellow) triggers first		Yellow LED ON; Green status LED ON
4	Player 2 presses after Player 1 has already won	✓ PASS	Player 2 LED stays OFF — lock confirmed
5	Player 1 presses after Player 2 has already won	✓ PASS	Player 1 LED stays OFF — lock confirmed
	Reset from Player 1 win state		All LEDs OFF; circuit reinitialised
	Reset from Player 2 win state		All LEDs OFF; circuit reinitialised
	Repeated trigger after clean reset		Priority locking resumed correctly

7. Discussion

7.1 Analysis of Results

All eight functional tests passed without exception across multiple repeated trials. The priority-locking mechanism worked reliably — no false triggering, no missed triggers, and no latch-up conditions were observed. The 1N4148 diode proved effective in both conducting the winner's output onto the status rail and in reverse-blocking to prevent back-driving of inactive modules.

7.2 Design Advantages

- **Simplicity** — Only standard, low-cost discrete components and a single IC type (NE555), ensuring high reliability.
- **Scalability** — Additional players require only per-module component replication; common rails are unchanged.
- **Supply Flexibility** — 555 Timer operates from 5 V to 15 V; compatible with USB, 9 V battery, or bench PSU.
- **Visual Clarity** — Separate player LEDs and a common status LED provide simultaneous, unambiguous feedback.
- **Robust Reset** — Pin 4 hardware override is independent of output/feedback state; guarantees clean reinitialisation.

7.3 Limitations and Future Improvements

- **No Audible Output** — A second 555 in monostable mode triggered by the status rail could produce a buzzer beep.
- **No Countdown Timer** — An astable 555 + decade counter could implement a response time window.
- **Breadboard Fragility** — A PCB design with an enclosure and long-reach cables would be needed for real deployment.
- **Battery Limitation** — A regulated DC adapter would be more suitable for larger multi-player configurations.

7.4 Learning Outcomes

- 555 Timer internal architecture: comparator thresholds, SR flip-flop, discharge transistor.
- Bistable multivibrator configuration without external timing components.
- Diode feedback networks for non-reciprocal signal routing on a shared bus.
- Resistor selection: Ohm's Law applied to LED current limiting and rail biasing.
- Systematic prototyping: phased assembly, incremental testing, and fault isolation.

8. Conclusion

The Fastest Finger First Circuit based on the NE555 Timer IC was successfully designed, assembled, and tested as part of the LICA End Semester Project. The circuit correctly identified the first of two players to press a trigger button, illuminated the corresponding LED, and reliably inhibited the other module — all within the near-instantaneous response time characteristic of hardware logic.

The priority-locking mechanism, implemented through a shared status rail and 1N4148 feedback diodes operating in bistable mode, proved robust across all eight functional test cases. The reset mechanism provided clean reinitialisation in every trial, and the common status LED gave the host clear round-level visibility.

Beyond demonstrating the practical utility of the 555 Timer, this project deepened the team's appreciation for systematic circuit design, where each component — from a 10 k Ω pull-up to a 0.6 V diode drop — plays an indispensable and precisely defined role. The insights gained in bistable IC design, feedback networks, and breadboard methodology will directly support more advanced integrated circuit work in future semesters.

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